

ABHINAND K

AI Engineer

Kozhikode, Kerala

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Summary

AI Engineer with 2 year of experience in Machine Learning, Large Language Models (LLMs), RAG pipelines, and Computer Vision. Skilled in building production-ready AI applications using Python, Transformers, Flask/FastAPI, and vector databases like Qdrant.

Skills

Programming: Python, SQL, HTML/CSS

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, Time Series Forecasting

LLM & Generative AI: Large Language Models (LLMs), RAG Pipelines, Prompt Engineering, Transformers, LangChain, Hugging Face, Ollama

Frameworks & APIs: Flask, FastAPI, REST APIs

Vector Databases: Qdrant, FAISS

Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, OpenCV, NLTK, Matplotlib, Seaborn

Tools: Docker, Git, Power BI, Tableau

Projects

AI-Powered Document Query Bot for Government Regulations (RAG)

Built an AI-powered document query bot for Kerala government employees, enabling instant access to official rules, regulations, and circulars using Retrieval-Augmented Generation (RAG) for accurate and up-to-date information retrieval.

Natural Language to SQL Query Conversion

Developed an LLM-powered system to convert natural language queries into SQL, allowing non-technical users to interact with databases efficiently. Implemented few-shot prompting to improve query accuracy and robustness.

PAN Masking (Computer Vision)

Developed an automated PAN document security system using YOLO to detect PAN numbers and sensitive fields, integrated with OpenCV-based image processing to accurately mask confidential information while preserving document integrity. Implemented preprocessing checks to skip already masked files.

Aadhaar Masking

Developed an automated Aadhaar document masking system using YOLO for detecting Aadhaar numbers and QR codes, combined with OpenCV image processing to protect sensitive information while maintaining document quality.

Form Line Removal (Computer Vision)

Built a computer vision solution using OpenCV to remove overlapping vertical lines from scanned bank forms, significantly improving OCR accuracy on handwritten and structured text documents.

Automatic Facial Attendance System

Developed a biometric face recognition-based attendance system including dataset creation, pre-processing, feature extraction, and classification, improving automation, accuracy, and security in attendance tracking.

Sign Language Identification using Computer Vision

Developed a Malayalam Sign Language Identification model that recognizes static and dynamic

gestures from real-time video streams and generates corresponding captions to assist the hearing and speech-impaired community.

Drug Review Classification using Machine Learning and Deep Learning Techniques

Built a sentiment classification system for drug reviews using manually labeled datasets and pre-trained transformer models such as BERT, SciBERT, and BioBERT, classifying reviews into positive, negative, and neutral sentiments.

AI-Powered Retail Forecasting System

Built a SKU-level demand forecasting and stock aging risk detection system using historical sales, inventory, and product master data. Applied time-series feature engineering and trained an XGBoost model to predict future demand and identify slow-moving or aging inventory.

Experience

AI Engineer — Ospyn Technologies (Aug 2024 – Present)

Working on Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Computer Vision projects, contributing to enterprise AI solutions and real-world intelligent systems development.

AI Intern — Ospyn Technologies (Feb 2024 – Jul 2024)

Worked on data science and machine learning projects with a focus on LLM applications, model development, and applied AI solutions in production-oriented environments.

NLP Research Intern — Applied NLP Research Lab, Digital University Kerala (Jul 2023 – Oct 2023)

Conducted sentiment classification on user-generated drug reviews using various Machine Learning and Deep Learning techniques, including pretrained transformer models and web-scraped datasets.

Data Science Intern — ICT Academy Kerala (Oct 2021 – Mar 2022)

Completed a Data Science and Analytics internship in collaboration with TCS iON at Technopark, Trivandrum, focusing on analytics workflows and practical data science applications.

Education

M.Sc. Computer Science (Data Analytics) — Digital University Kerala (2022 – 2024)
CGPA: 7.89

B.Sc. Computer Science — College of Applied Science Vazhakkad, (2018 – 2021)
CGPA: 6.6

Higher Secondary Education — Govt. HSS Kuttikkattoor, Kozhikode (2016 – 2018)
CGPA: 7.9

Certifications

Databricks Certified Generative AI Engineer Associate

Certification focused on Generative AI, Large Language Models, and enterprise-level AI solution development using modern AI frameworks and tools.

Publications

Sentiment Analysis of Drug Reviews

Published research on classifying drug reviews using BERT, SciBERT, and BioBERT embeddings with machine learning and deep learning models, evaluated using precision, recall, and F1-score.

URL: <https://arxiv.org/abs/2404.13057>